PRE-STUDY · FINAL PROJECT

Eight Problems, One Pipeline

What machine learning adds beyond classical statistics

Financial Machine Learning (QRM3007) · Fall 2026 · Prof. Keehong Rhee

Read this before your team chooses a project. For each problem: the classical method, what machine learning adds, and the trap that can erase the gain.

This pre-study makes three points

Read it before your team chooses one of the eight projects

1

Every project starts from a classical method

Finance already attacks each problem with statistics. That method is your baseline.

2

Machine learning gains in four specific situations

Nonlinear patterns, many inputs, text or networks, and decisions at scale.

3

The gain counts only if it survives out of sample

Honest testing turns a promising model into a result you can report.

Learn the classical method first; then show what machine learning adds.

Statistics explains; machine learning predicts

Breiman (2001) called these the two cultures of statistical modelling

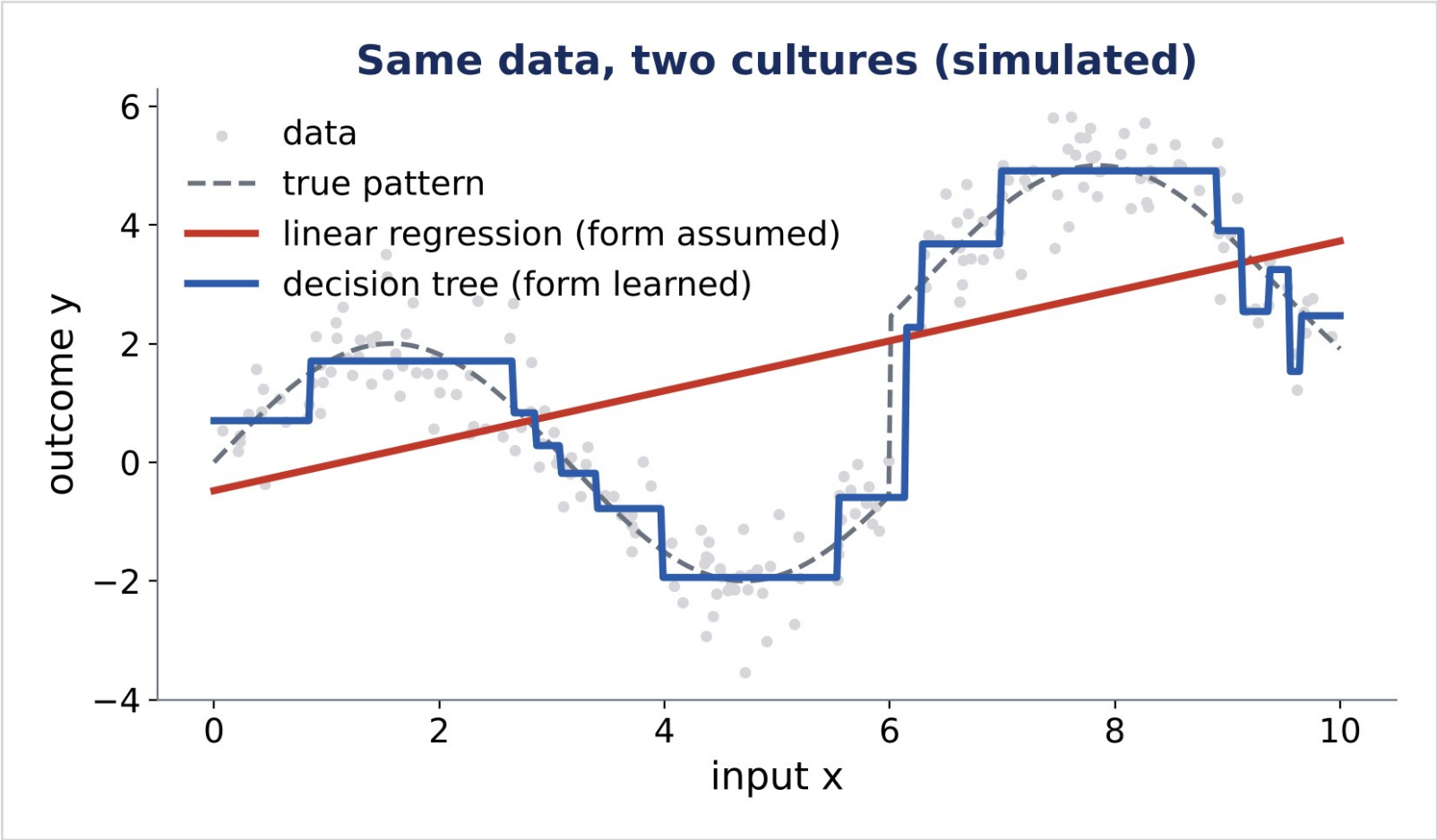
	Classical statistics	Machine learning
Goal	Explain how inputs affect the outcome	Predict new cases well
Model form	Chosen by the analyst	Learned from the data
Judged by	Fit and p-values in the sample	Error on unseen data
Strength	Clear; works with little data	Finds complex patterns
Weakness	Misses what it did not assume	Overfits without honest tests

A learned form follows patterns a line misses

Simulated data: a straight line versus a decision tree

Read the chart

- Grey dots: the data
- Dashed: the true pattern
- Red: form chosen in advance
- Blue: form learned from data



Machine learning gains in four situations

Each final project sits in at least one of them



Nonlinear patterns

P1 · P4 · P7

Effects that bend, jump or interact

risk jumps near the limit



Many inputs

P2 · P3 · P6

Dozens of features or assets at once

30 card features



New data types

P5 · P8

Text and networks, not tables

releases, payment graphs



Decisions at scale

P2 · P6 · P8

Rank, filter and alert many cases

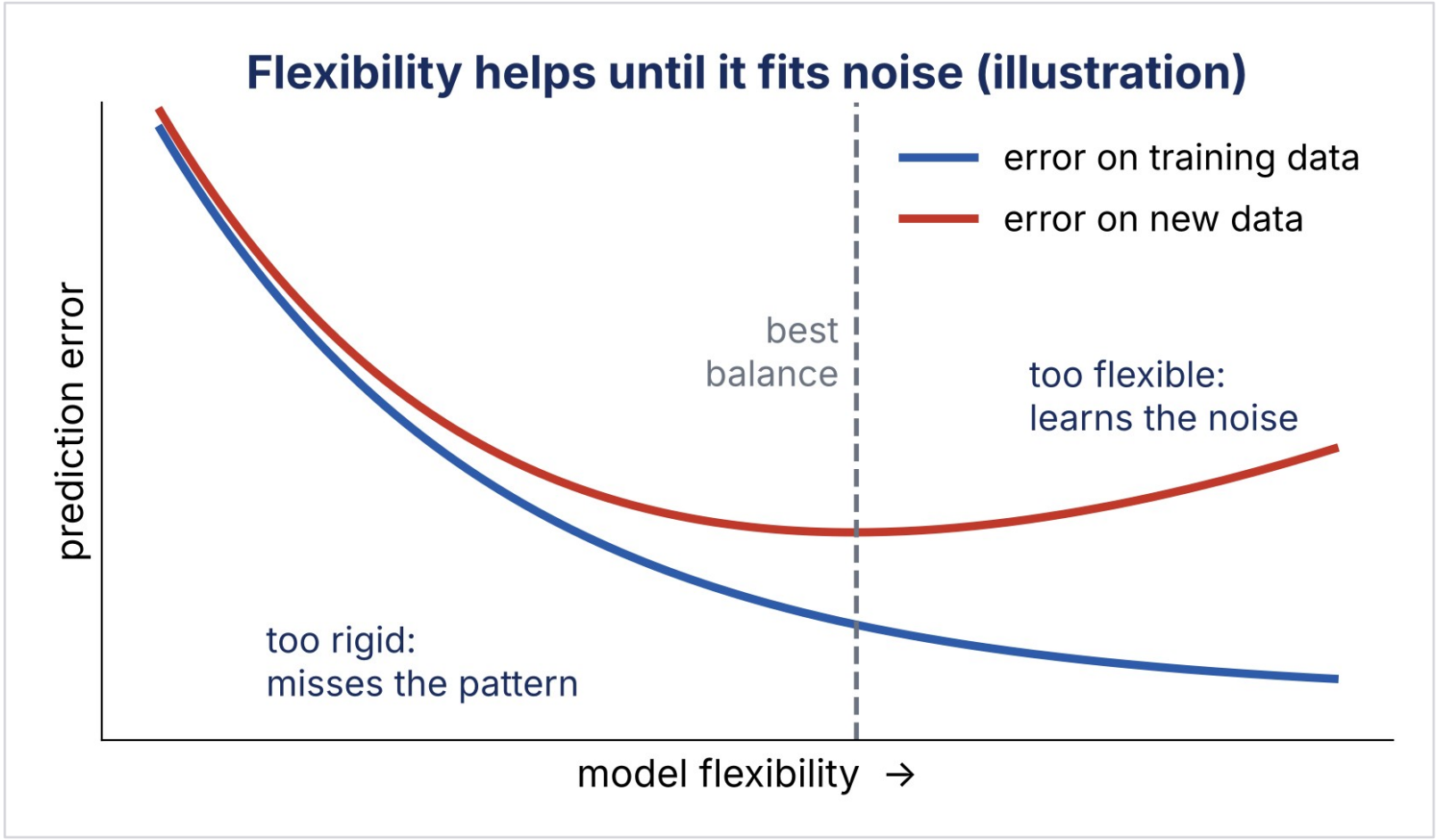
top 100 alerts a day

Flexibility has a price: overfitting

A flexible model can fit the training data and still fail on new data

Read the chart

- Blue keeps falling: training error
- Red turns up: new-data error
- Honest tests measure the red curve



Eight projects, eight classical baselines

Each tile: the method finance used first, and the machine learning challenger

	CLASSICAL		MACHINE LEARNING
P1 Meta-labeling	Trading rule + t-test	>	Random forest on signals
P2 Card fraud	Rules, z-score limits	>	LOF, Isolation Forest
P3 Portfolio	Markowitz mean-variance	>	HRP clustering
P4 Recession	Probit on yield spread	>	Boosting + SHAP
P5 Sentiment	Dictionary word counts	>	FinBERT model
P6 Recommender	Popularity, basket rules	>	Matrix factorisation
P7 Credit	Logistic scorecard	>	Boosting + fairness
P8 AML	Threshold rules	>	Graph neural network

01

MARKETS

Trading Signals and Portfolios

P1 Meta-labeling a trend signal · P3 Hierarchical Risk Parity

A fixed trading rule treats every signal alike

Moving-average crossover on Bitcoin dollar bars, 2021–2025

When the fast average crosses the slow one, buy. Are all crossovers equally good?

CLASSICAL

Trading rule + t-test

- Buy on every crossover
- Test the average return once

Blind spot: every signal treated the same

MACHINE LEARNING

Meta-labeling with a random forest

- Label each signal by its outcome
- Learn when signals work from volatility, volume, trend

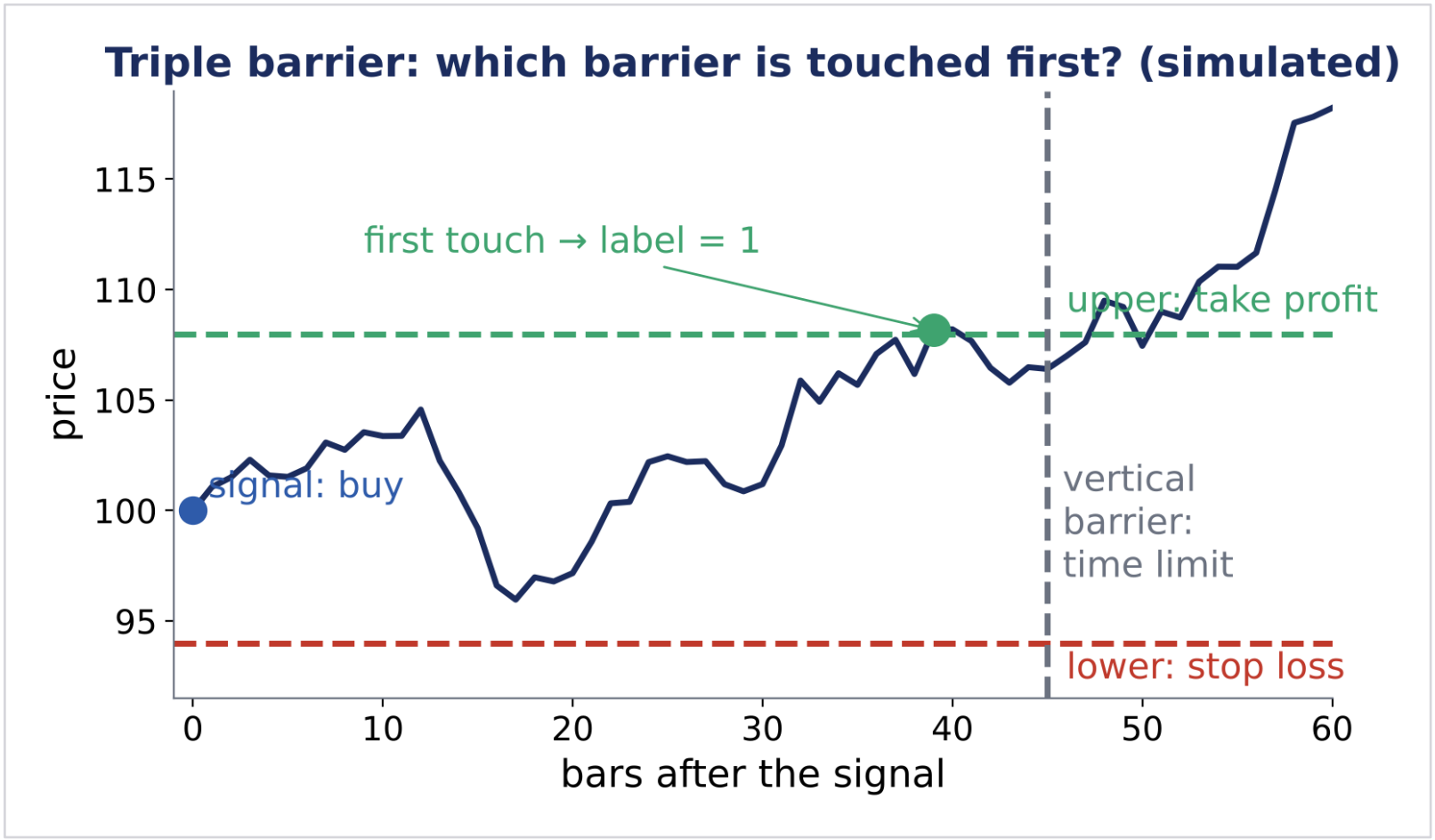
Gain: skip weak signals, precision rises

The triple barrier labels each signal by its outcome

Take profit, stop loss or time limit: whichever is touched first decides

Read the chart

- Blue dot: the rule says buy
- Green hit first: a good signal (1)
- Red first or time out: a bad signal (0)
- Now a model can learn which signals win



Trading-rule profits have faded under honest tests

Why the meta-model must be tested more strictly than the rule

EPISODE

1992

Brock, Lakonishok & LeBaron: moving-average rules looked profitable on the Dow, 1897–1986

1999

Tests that correct for data snooping found the edge much weaker after 1986

HONEST-TEST CATCH

- Labels overlap in time: purge and embargo
- Count every rule you tried; report the Deflated Sharpe Ratio
- Include trading costs

Mean-variance inverts a noisy covariance matrix

About 20 ETFs across asset classes, 2005–2025

How should we split money across 20 funds so that risk is as low as possible?

CLASSICAL

Markowitz mean-variance

- Estimate the covariance matrix
- Invert it to get the weights

Blind spot: small errors become wild weights

MACHINE LEARNING

Hierarchical Risk Parity

- Cluster similar assets into a tree
- Split risk down the tree, no inversion

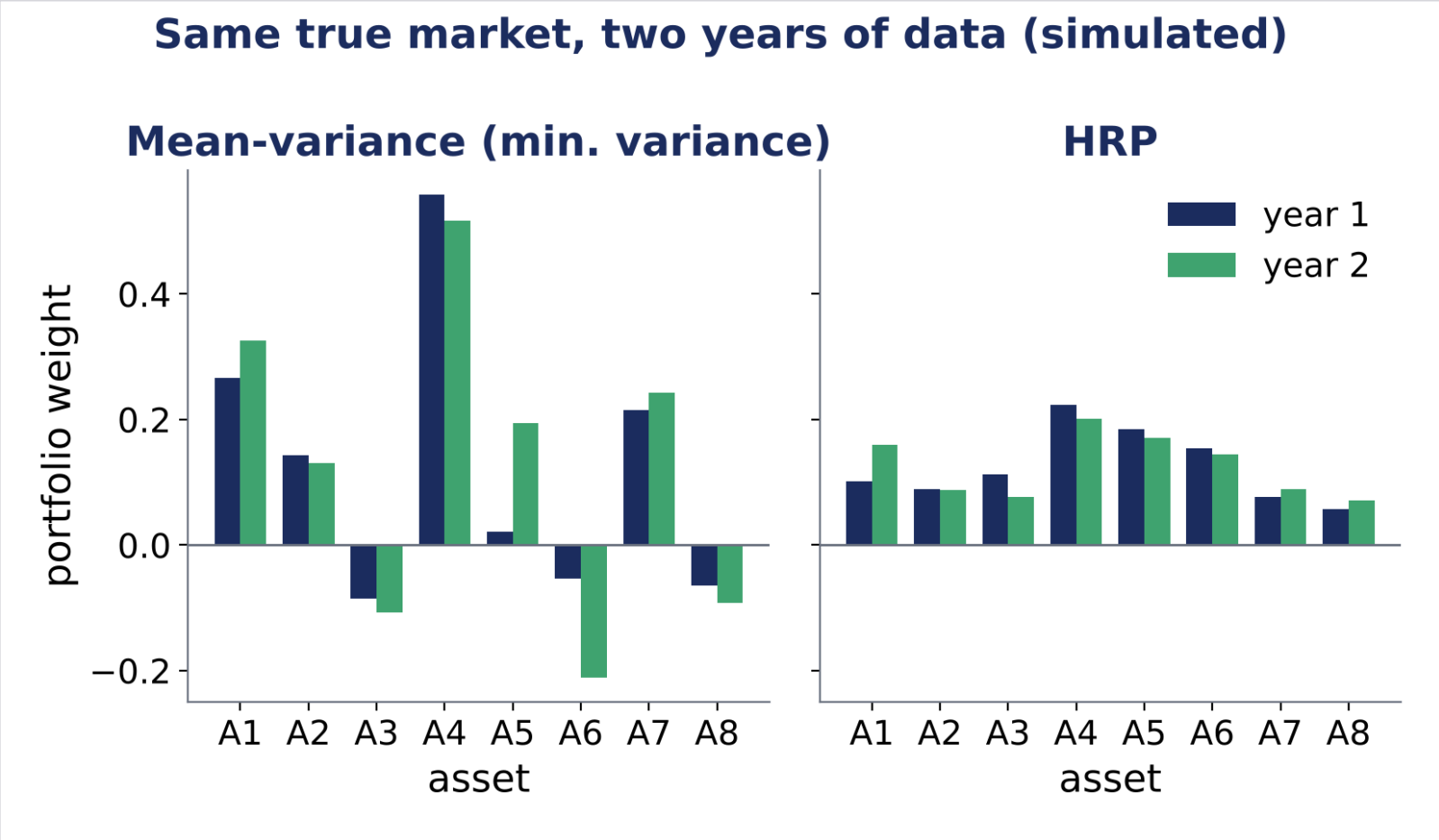
Gain: stable, diversified weights

Same market, another year, very different weights

Simulated returns from one true covariance; only the sample changes

Read the chart

- Left: minimum-variance weights jump and turn negative
- Right: HRP weights barely move
- The market did not change; only the noise did



Simple 1/N is the benchmark to beat

Four decades of evidence on optimised portfolios out of sample

EPISODE

1952
Markowitz publishes portfolio selection: the start of modern portfolio theory

2009
DeMiguel, Garlappi & Uppal: 14 optimised portfolios fail to beat 1/N consistently out of sample

2016
López de Prado proposes HRP

HONEST-TEST CATCH

- Beat 1/N after costs, not only mean-variance
- Fix the 252-day look-back before testing
- Enter each ETF only after it has history

02

MACRO AND TEXT

Recessions and Earnings Releases

P4 Recession probability · P5 Earnings-release sentiment

One yield-curve signal has called most recessions

Monthly FRED data, 1970–2025

What is the chance of a U.S. recession within the next 12 months?

CLASSICAL

Probit on the yield spread

- One signal: the term spread
- Inverted curve raises the probability

Blind spot: ignores how signals combine

MACHINE LEARNING

Gradient boosting + SHAP

- Many indicators at once
- Finds thresholds and combinations

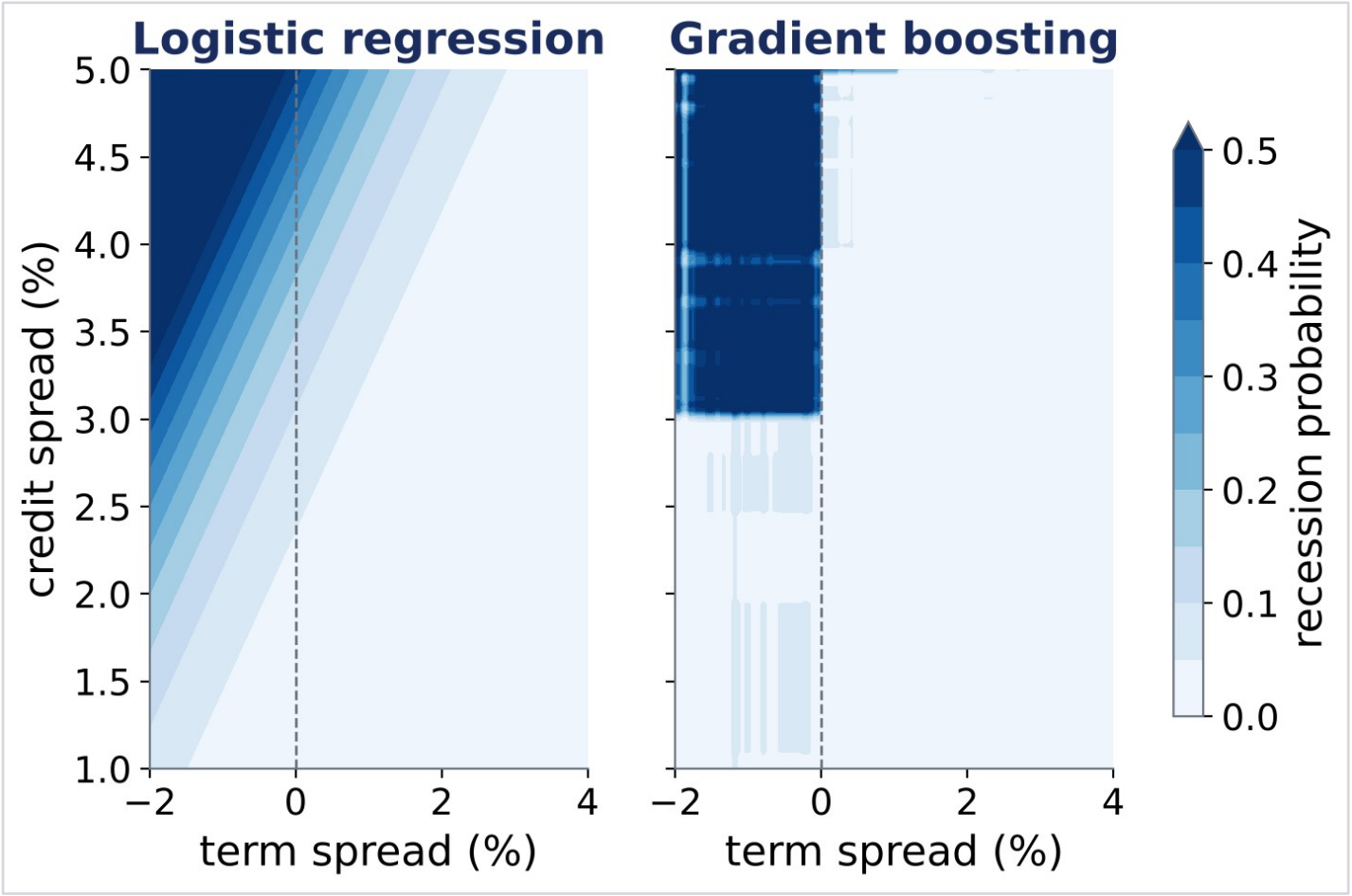
Gain: interactions, explained month by month

Boosting finds that signals matter together

Simulated: risk is high only when the curve is inverted and credit is stressed

Read the chart

- Darker means higher recession probability
- Left: logistic can only tilt a line
- Right: boosting finds the corner where both signals fire



A famous signal can still mislead

The 2022 inversion and the limits of eight recessions

EPISODE

1998

Estrella & Mishkin show the term spread predicts recessions a year ahead

2022

The curve inverts and stays inverted for about two years

2024

No recession has been declared: one signal alone can mislead

HONEST-TEST CATCH

- Only about eight recessions since 1970: keep models small
- Use data as published at the time
- Never use the recession indicator as a feature

Counting words misreads financial language

Earnings press releases in SEC 8-K filings, 2015–2025

Does the tone of an earnings release predict the stock's reaction?

CLASSICAL

Dictionary word counts

- Count positive and negative words
- Tone = difference / total words

Blind spot: words read without context

MACHINE LEARNING

FinBERT language model

- Reads each sentence as a whole
- Trained further on financial text

Gain: negation, contrast and finance vocabulary

Context changes what a word means

Word counting versus a language model that reads the whole sentence

SENTENCE	WORD COUNT	IN CONTEXT
“Costs fell 12% from last year.” trigger words: costs, fell	Negative	Positive
“We saw no decline in demand.” trigger words: decline	Negative	Positive
“Revenue beat forecasts, but guidance was cut.” trigger words: beat	Positive	Negative
“Tax liability was in line with plan.” trigger words: tax, liability	Negative	Neutral

Finance needs its own vocabulary

From finance word lists to finance language models

EPISODE

2011

Loughran & McDonald: almost three-quarters of a general dictionary's negative words are not negative in finance

2019

FinBERT: a BERT language model trained further on financial text

HONEST-TEST CATCH

- Tone and earnings surprise move together: control for surprise
- Releases after 4 pm count from the next trading day

03

CUSTOMERS

Recommendation and Credit

P6 Next-best-product recommender · P7 Credit scoring with fairness

Basket rules give everyone the same advice

Santander bank: 24 products, monthly holdings, 2015–2016

Which product will this bank customer add next month?

CLASSICAL

Popularity and basket rules

- Recommend the best sellers
- Customers with A often buy B

Blind spot: same advice for everyone

MACHINE LEARNING

Matrix factorisation + ranker

- Hidden taste factors per customer
- Adds age, income and tenure

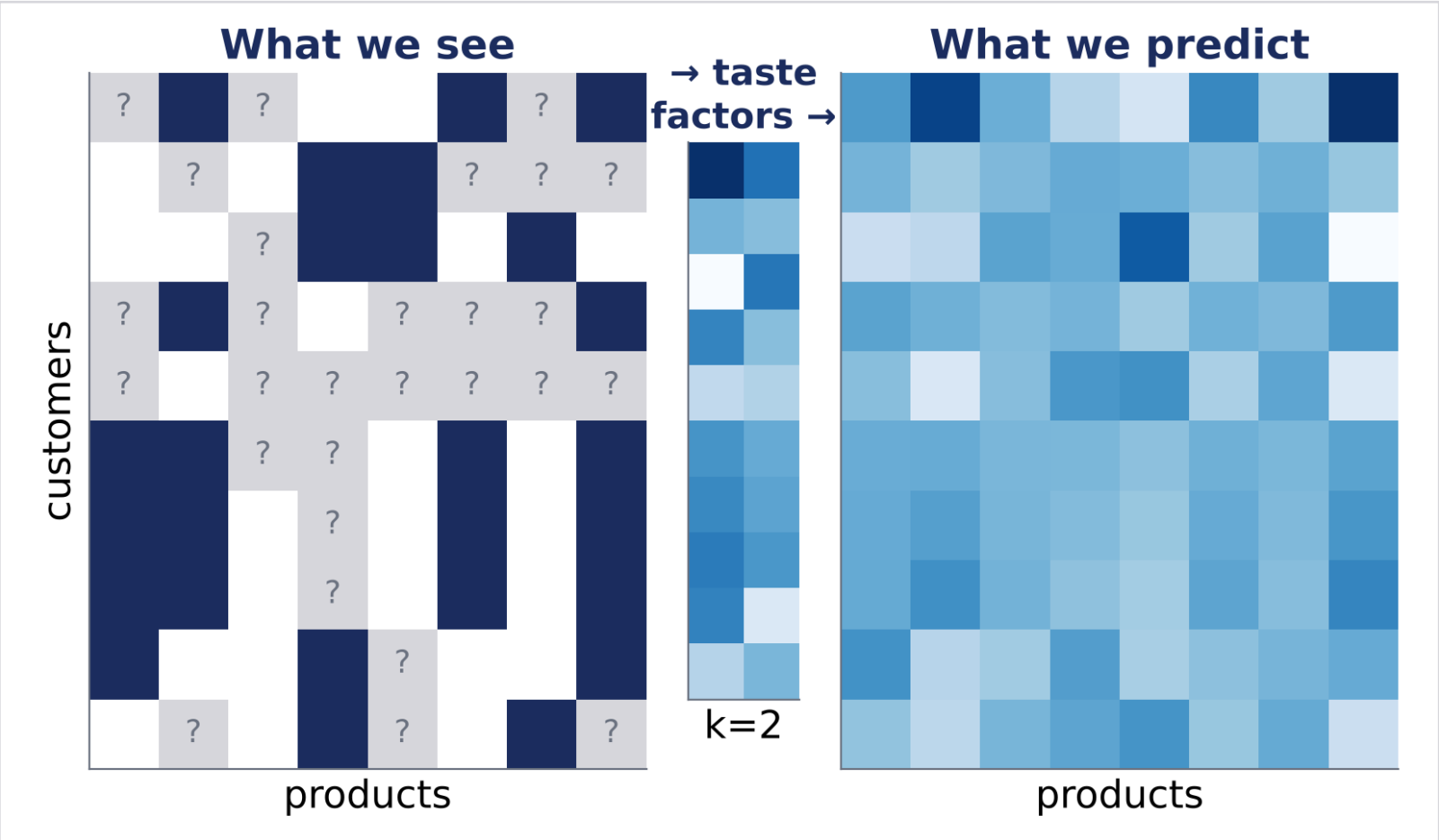
Gain: personal advice, at scale

Matrix factorisation fills in the blanks

Customers and products get a few hidden taste factors; together they predict the blanks

Read the chart

- Navy: product held
- ?: unknown — most cells
- Middle: two taste factors per customer
- Right: a score for every pair



A prize made matrix factorisation famous

What the Netflix Prize taught recommenders

EPISODE

2006

Netflix offers USD 1 million for a 10% better forecast of ratings

2009

The prize is won; matrix factorisation is at the heart of the winning methods

HONEST-TEST CATCH

- Train only on months before the test month
- Check new customers with no history
- A product must suit the client

Credit scorecards add up a few fixed weights

30,000 credit-card clients in Taiwan, 2005

Will this client default next month, and can we explain the decision?

CLASSICAL

Z-score and logistic scorecard

- A few ratios with fixed weights
- Transparent and trusted by regulators

Blind spot: one slope for every client

MACHINE LEARNING

Gradient boosting + fairness checks

- Learns thresholds and interactions
- Gives each client three reasons

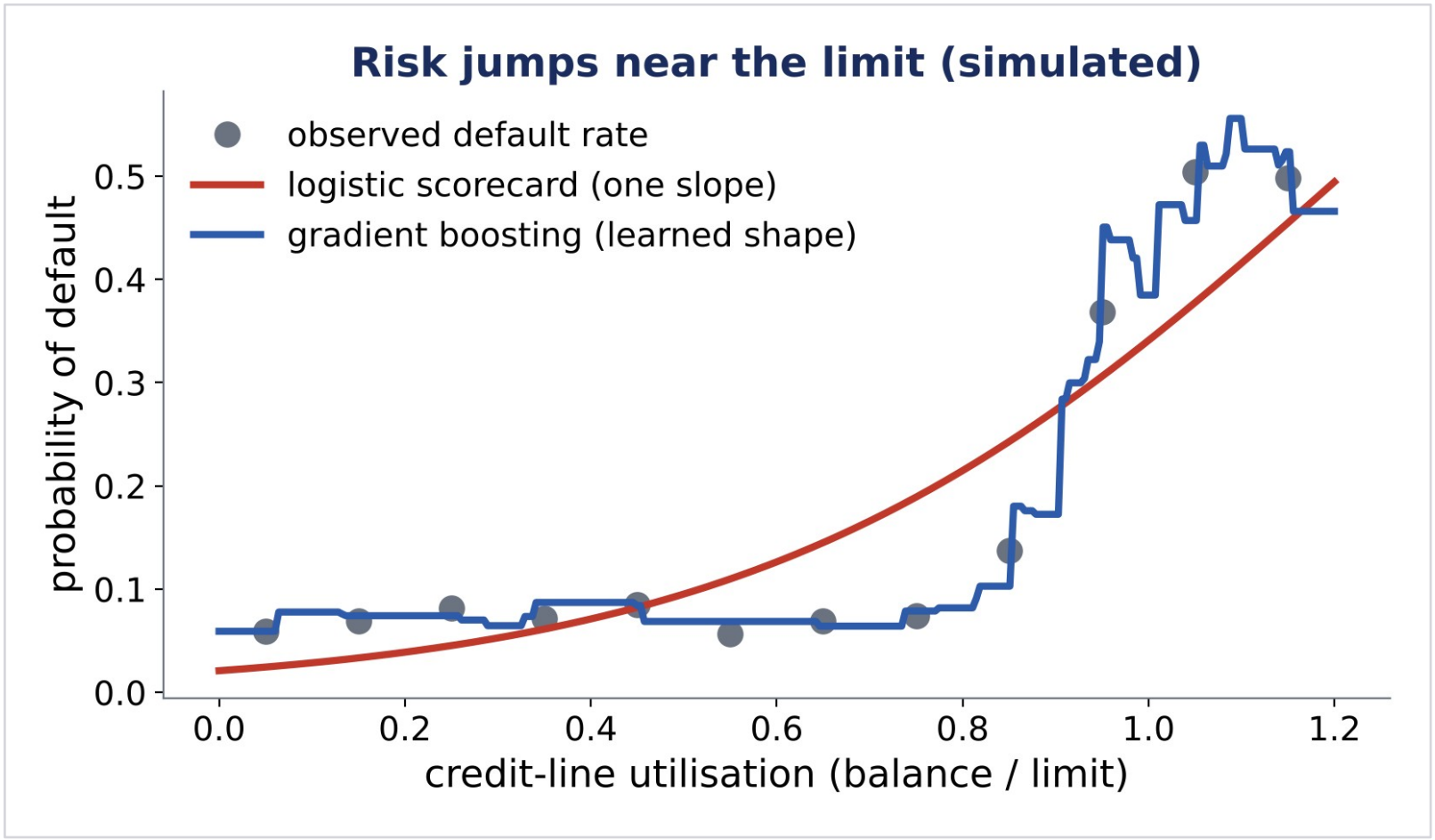
Gain: sharper risk ranking with reasons

Default risk jumps near the credit limit

Simulated: one slope smooths over the jump that boosting finds

Read the chart

- Grey dots: observed default rates
- Red: one logistic slope
- Blue: shape learned from data
- The jump sits near 90% utilisation



Credit decisions must be fair and explained

From the Z-score to a credit-limit controversy

EPISODE

1968
Altman's Z-score adds up five financial ratios

2019
Users allege a new credit card gives women lower limits

2021
New York's regulator finds no unlawful discrimination but criticises the lack of transparency

HONEST-TEST CATCH

- Gains over a good scorecard can be small: measure them
- Dropping sex or age does not remove proxies

04

ANOMALIES

Card Fraud and Money Laundering

P2 Card fraud detection · P8 Money-laundering detection on a graph

Rules check one feature at a time

European card transactions over two days in 2013

Which card transactions should the team investigate today?

CLASSICAL

Rules and z-score limits

- Flag amounts far from normal
- Check each feature alone

Blind spot: fraud that is normal on every feature

MACHINE LEARNING

Unsupervised detectors

- LOF, Isolation Forest, autoencoder
- Look at all features together

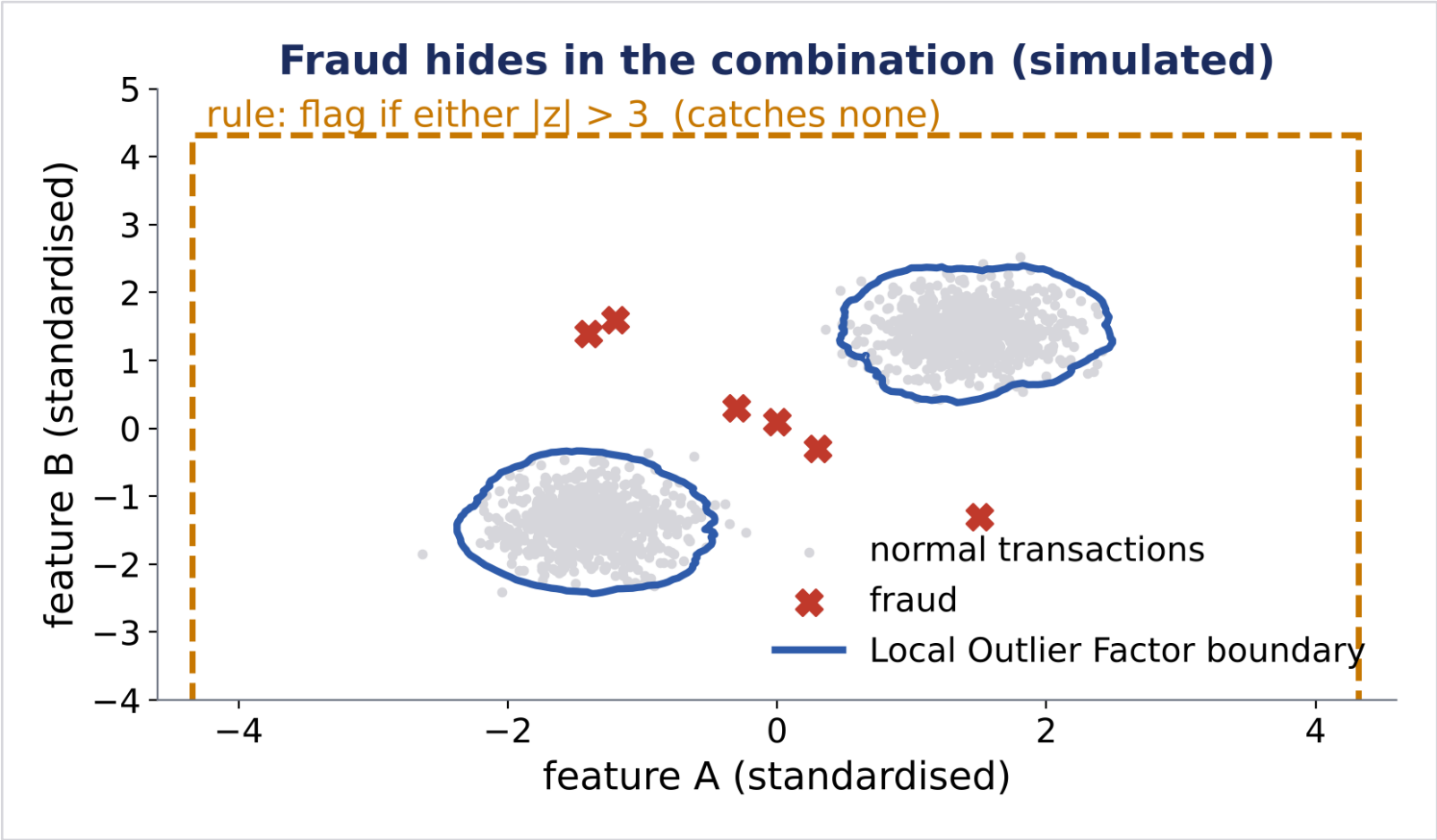
Gain: rank the strangest first, no labels

Fraud hides in combinations of features

Simulated: each fraud looks normal on every feature alone

Read the chart

- Amber box: z-score rule, catches none
- Blue: density boundary (LOF)
- All six frauds fall outside it
- No fraud labels were used



Accuracy is the wrong score for rare fraud

0.17% fraud means a lazy model looks almost perfect

EPISODE

2013

European cardholders: 284,807 transactions over two days, 492 frauds (0.17%)

Lesson

Saying 'never fraud' scores 99.83% accuracy and catches nothing

HONEST-TEST CATCH

- Judge by PR-AUC and precision@100, not accuracy
- Split in time; never tune on the test period

AML rules inspect one payment at a time

Elliptic Bitcoin graph: 203,769 transactions over 49 time steps

Is this Bitcoin transaction part of a scam, ransomware or dark-market flow?

CLASSICAL

Threshold rules

- Amounts and known patterns
- One payment at a time

Blind spot: ignores who pays whom

MACHINE LEARNING

Graph neural network

- Passes evidence along payment links
- Combine with a random forest

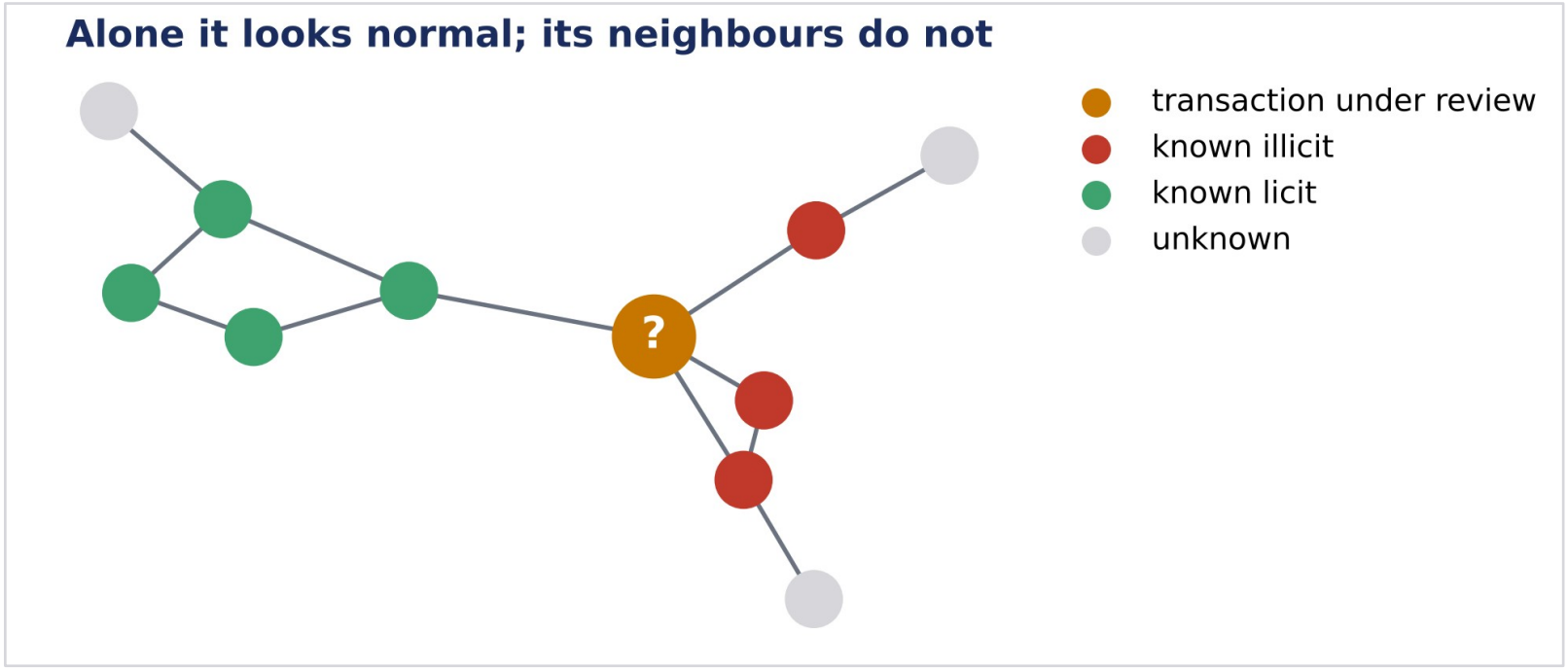
Gain: evidence from neighbours

Who you trade with is evidence

A transaction that looks normal can sit among known illicit ones

Read the chart

- Its own features say normal
- Three of its four neighbours are illicit
- Rules that read one row miss this



Models break when the market changes

The Elliptic data and the dark-market shutdown

EPISODE

2019

Weber et al. release the Elliptic data: 203,769 labelled-or-unknown transactions

Step 43

A dark market closes; afterwards no model reliably detects illicit flows

HONEST-TEST CATCH

- A random forest on features was a strong baseline
- Test on later time steps and show where it breaks

Each project has an edge and a trap

Where machine learning gains over the classical method, and what can erase the gain

	ML EDGE	TRAP TO AVOID
P1 Meta-labeling	Context picks the signals to take	Overlapping labels, no costs
P2 Card fraud	Joint patterns, no labels needed	Judging by accuracy
P3 Portfolio	No inversion, stable weights	Not beating 1/N after costs
P4 Recession	Interactions, explained by SHAP	Few events, revised data
P5 Sentiment	Reads words in context	Ignoring the earnings surprise
P6 Recommender	Personal hidden tastes	Using future months
P7 Credit	Thresholds and reasons per client	Proxy bias, small gains
P8 AML	Evidence from neighbours	Market change after training

SUMMARY

The classical model is the benchmark, not the enemy

Every project reports both; the out-of-sample test decides

Classical tends to win when

- Data are few or the signal is weak
- You must prove an effect
- Regulation fixes the model's form

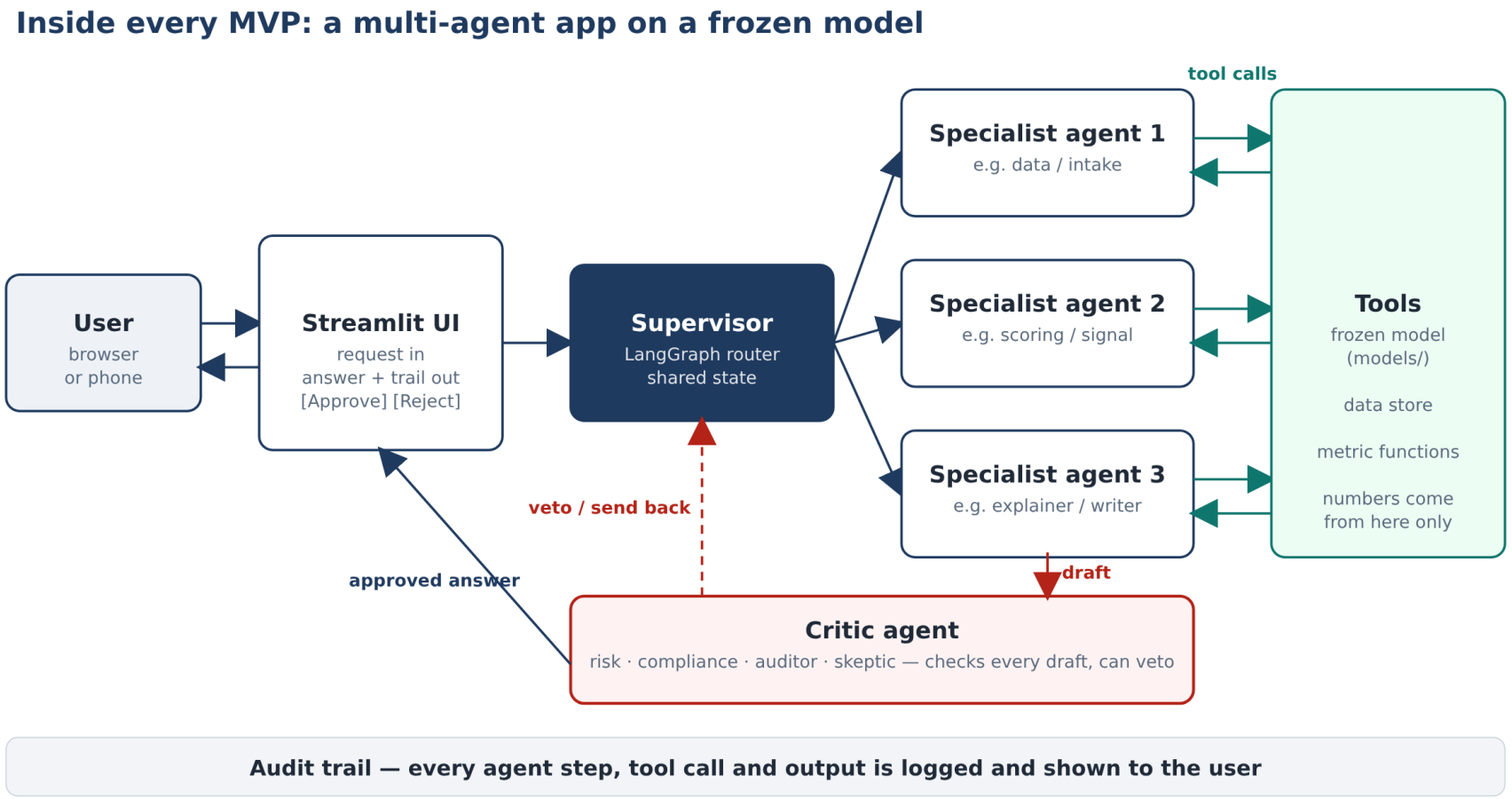
Machine learning tends to win when

- Patterns bend or interact
- Inputs are many, text or networks
- Decisions are many: rank, filter, alert

If ML does not beat the baseline honestly, report that. It can still earn full marks.

Every project ends as an app run by agents

Your frozen model becomes a tool the agents call; a critic agent can say no



DISCUSSION

Four questions help your team choose

Discuss them before the end of Week 9

Q1	Which data interests you most: prices, text, customers or networks?	**Interest**
Q2	Can you state the classical method for that project in one sentence?	**Baseline**
Q3	What would an honest failure look like in that project?	**Test**
Q4	Who would use your app, and what would they approve?	**Product**

Baseline first, challenger second, honest test last

The order in which every final project is built and judged

1. Baseline

Start from the classical method finance already trusts

2. Challenger

Add machine learning where the four situations apply

3. Honest test

Time-ordered splits, purging and costs decide what you report

Next: choose your project and team (M0, end of Week 9).

A

APPENDIX

Key Formulas at a Glance

One core formula per project, for reference while you build

Each project rests on one core formula

Read the label first; the formula states the same idea precisely

P1 Meta-labeling
act only on likely winners

$$z_i = \mathbf{1}\left[\text{side}_i \cdot r_i > 0\right], \qquad \text{trade only if } \hat{P}\left(z_i = 1 \mid x_i\right) > 0.5$$

P2 Card fraud
isolation score

$$s(x, n) = 2^{-E[h(x)]/c(n)}$$

P3 Portfolio
correlation distance and split

$$d_{ij} = \sqrt{\tfrac{1}{2}\left(1 - \rho_{ij}\right)}, \qquad \alpha = 1 - \frac{V_1}{V_1 + V_2}$$

P4 Recession
SHAP adds up to the forecast

$$f(x_t) = \phi_0 + \sum_{j=1}^{M} \phi_j(x_t)$$

Each project rests on one core formula

Read the label first; the formula states the same idea precisely

P5 Sentiment
event-study regression

$$CAR_i = a + b \cdot \mathrm{Sent}_i + c \cdot \mathrm{Surp}_i + \varepsilon_i$$

P6 Recommender
taste factors meet

$$\hat{r}_{ui} = p_u^\top q_i$$

P7 Credit
equal-opportunity gap

$$\Delta_{\mathrm{EO}} = \left| \mathrm{TPR}_A - \mathrm{TPR}_B \right|$$

P8 AML
graph convolution layer

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2} \tilde{A} \, \tilde{D}^{-1/2} H^{(l)} W^{(l)} \right)$$